

Luis Diego Montero Vargas

Data Analyst | Data Scientist | Business Intelligence | Machine Learning

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Profile

Big Data and Information Technology professional with over 5 years of experience in high-demand technological environments and an active focus on data analysis, data science, and business intelligence. Hands-on experience across the full data lifecycle: collection, cleaning, transformation (ETL), modeling, statistical analysis, Machine Learning, and visualization. Builds interactive dashboards and executive reports in Power BI (DAX, Power Query, KPIs) and advanced Excel; performs analysis and modeling with Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn) and R; queries and manages SQL, Oracle, and NoSQL databases. Projects publicly deployed on Streamlit with integration of external APIs (NASA). Certified in Data Science, Data Mining, and agile methodologies (Scrum). Bilingual Spanish/English profile, results-oriented, with the ability to communicate findings clearly to both technical and non-technical audiences.

Technical Skills

Languages and analysis

- **Python:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Folium, Streamlit — exploratory analysis, statistical modeling, and Machine Learning
- **R / RStudio:** statistical analysis, data visualization, and modeling
- **SQL / Oracle:** advanced queries, joins, subqueries, data transformation and extraction (ETL)

Business Intelligence and visualization

- **Power BI:** executive dashboard design, KPI and performance indicator construction, custom DAX measures, Power Query, data modeling, geographic visualizations, and customer segmentation
- **Advanced Excel:** pivot tables, complex formulas, data modeling, charts, and executive reports

Machine Learning and data science

- Predictive modeling with Python (Scikit-learn), pattern analysis, probability distributions (binomial, Poisson, normal), descriptive and inferential statistics
- External API integration (NASA API), geospatial visualization (Folium), data application deployment on Streamlit

Databases and infrastructure

- Relational databases (SQL, Oracle) and non-relational (NoSQL)
- ETL processes, data cleaning and consolidation from multiple sources
- Version control: Git / GitHub — collaborative projects and technical documentation

Environments and tools

- Jupyter Notebook, RStudio, Visual Studio Code, PyCharm, Visual Studio 2022, Streamlit
- Agile methodologies: Scrum (SFC certified)
- Languages: Spanish (native) — English (intermediate B1: listening 8/10, speaking 7/10, reading 7/10)

Data Science and Data Analysis Projects

Energy Consumption Prediction in Costa Rica (Python, Machine Learning, NASA API, Folium, Pandas, Streamlit) 2026

- Developed an analytical and predictive model using real data from ARESEP and environmental variables obtained through the NASA API to identify national energy consumption patterns.
- Implemented a full pipeline: external API data ingestion, cleaning, statistical analysis, Machine Learning modeling, and interactive geographic visualization with Folium.
- Publicly deployed on Streamlit: [see application in portfolio](#)

Business Intelligence Dashboard — Bicycle Sales (Power BI, DAX, Power Query, Excel)

2026

- Collected, cleaned, and modeled 2016-2018 sales data (California and Texas) using Power Query to ensure data integrity and quality.
- Designed an executive dashboard with commercial KPIs, customer segmentation, geographic visualizations, and sales trend and profitability analysis.
- Built custom DAX measures for dynamic analysis of business indicators to support strategic decision-making.

Election Results Simulation (Python, Power BI, statistical analysis)

2025

- Handled and simulated real data from Costa Rica's electoral roll: over 3 million voters across all of the country's provinces, cantons, districts, and polling stations.
- Performed statistical analysis of participation patterns and behaviors; visualized results in Power BI with maps, KPIs, and candidate outcome projections for both technical and non-technical audiences.

Statistical Analysis — Probability Distributions (Python, Pandas, Matplotlib, Seaborn)

2025

- Applied binomial, Poisson, and normal distributions to a dataset of over 3,000 records; performed inferential analysis and produced reports with clear visualizations to communicate findings effectively.

Movie Insights — Collaborative Data Analysis (OOP) (Python, Pandas, Streamlit, Matplotlib, Seaborn, GitHub)

2025

- Collaborative TMDB data analysis project (2020-2025) with object-oriented architecture, version control on GitHub, and deployment on Streamlit.
- Professional technical documentation: structured README, dependency management, and collaborative development best practices.

Professional Experience

Technical Support Specialist

Jun 2019 – Feb 2024

Universidad Latina | Universidad Americana

- Provided multichannel technical support (in person, remote, and by phone) for hardware, software, and network connectivity in a high-demand multi-campus university environment.
- Managed users in Windows Active Directory and Office 365: provisioning, deactivation, permissions, licenses, and domain administration.
- Documented cases in a ticketing system, maintained IT asset inventory, and produced incident reports for service indicator tracking.
- Implemented preventive and corrective maintenance plans and created technical guides for end users.

Projection and Sound Technician

Dec 2013 – Dec 2018

Cinépolis

- Administered Linux servers for digital projection operations and performed corrective maintenance on technological equipment.
- Coordinated screening schedules and managed advertising slots according to box office performance indicators.

Certifications

- **Introduction to Data Science** — MICITT-CECI / Cisco Networking Academy *Jul 2026*
- **Scrum Fundamentals Certified (SFC)** — SCRUMstudy *Nov 2025*
- **Introduction to Data Mining** — Pontificia Universidad Católica de Chile, via Coursera *Dec 2022*

Education

Big Data Program (in progress) *Jan 2023 – Present*
Colegio Universitario de Cartago (CUC)

High School Diploma *Dec 2018*
Colegio Vicente Lachner Sandoval